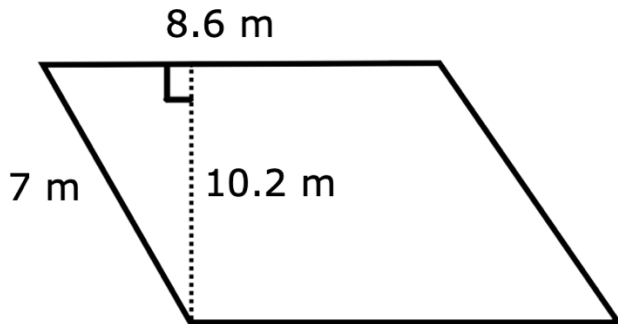


A parallelogram with dimensions labeled is shown below for questions 1 – 2.



1. Which dimension is not needed to find the area of the parallelogram? Explain.
Area = base $\cdot$ height. The length of the slanted side is neither the base nor height, so it is not needed.

2. Fill in the blanks to find the area of the parallelogram.

$$\text{Area of parallelogram} = b \cdot h$$

$$\text{Area} = \underline{8.6} \cdot \underline{10.2}$$

$$\text{Area} = \underline{87.72} \text{ in.}^2$$

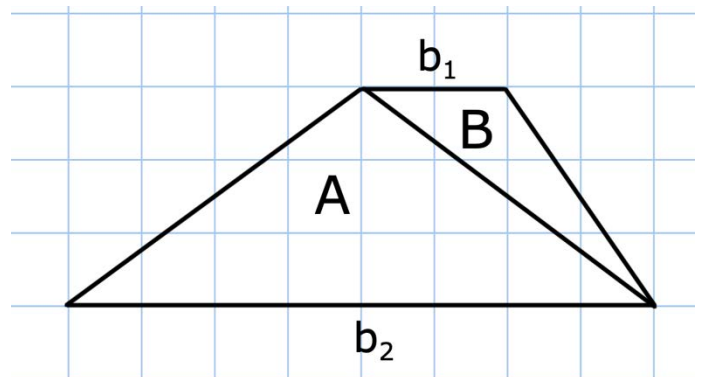
A trapezoid can be decomposed into two triangles with the same height. Use the diagram below for questions 3 – 6.

3. The base of triangle A, b_2 , is 8 units and its corresponding height is 3 units. Therefore, the area of triangle A is 12.

$$\frac{1}{2} \cdot 8 \cdot 3 = 12$$

4. The base of triangle B, b_2 , is 2 units and its corresponding height is 3 units. Therefore, the area of triangle B is 3.

$$\frac{1}{2} \cdot 2 \cdot 3 = 3$$



5. The area of the trapezoid is $\frac{15}{12 + 3 = 15}$ units².

6. Use the formula for area of a trapezoid to verify your solution is correct.

$$\text{Area} = \frac{1}{2}h(b_1 + b_2).$$

$$\text{Area} = \frac{1}{2} \cdot 3 \cdot (2 + 8)$$

$$\text{Area} = 15 \text{ units}^2$$

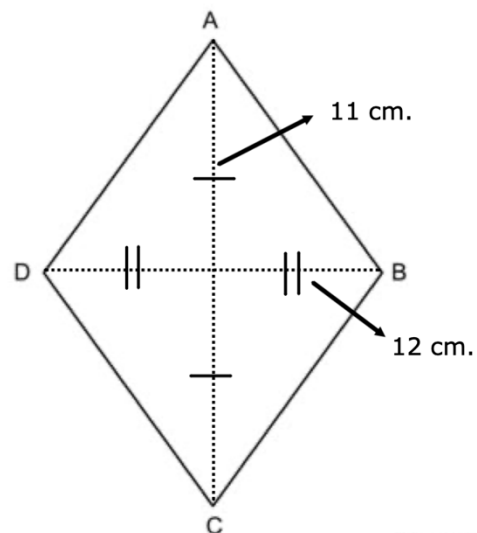
Rhombus $ABCD$ is shown. Diagonal BD is 12 centimeters and diagonal AC is 11 centimeters. Use the diagram below for questions 7 – 10.

7. The base of triangle ABC is 11 cm. and its corresponding height is 6 units. Therefore, the area of triangle ADC is 33 .

$$\frac{1}{2} \cdot 11 \cdot 6 = 33$$

8. The base of triangle ADC is 11 cm. and its corresponding height is 6 units. Therefore, the area of triangle ADC is 33 .

$$\frac{1}{2} \cdot 11 \cdot 6 = 33$$



9. The area of the trapezoid is $\frac{66}{33 + 33 = 66}$ cm².

10. Use the formula for area of a rhombus to verify your solution is correct.

$$\text{Area} = \frac{1}{2}(d_1)(d_2).$$

$$\text{Area} = \frac{1}{2}(\underline{11})(\underline{12})$$

$$\text{Area} = 66 \text{ units}^2$$